Alex's To-Do List

This document is the single running punch list of things only you can do — configuration steps that need your hands on a credential, data entry that needs your domain judgment, or reference lookups that help steer the next round of development. I'll maintain it as items land or become obsolete.

How to use this file:

- Items are grouped by type, not by priority. The **Priority** line inside each item is what ranks them.
- Each item says what to do, why it matters, and how to verify it's done.
- **Don't delete completed items** — move them to the "Done" section at the bottom with a short note on what it unlocked.

Last updated: 2026-04-30 (Gate L4 / L5 / L6 i18n shipped — French content + UI is human-review-ready, German + Spanish are MT-staged post-ABQ. **§2.11 added below for the language-comparison approval pass (143 FR rows in `writer_reviewed` waiting for you).** Also a demo-shaped coordinator-dashboard seed landed in `data/seed/coordinator-demo/` so the panels render non-empty even before §2.1 / §2.7 entries land — see §2.1 update.)

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1. Configuration

§1.1–1.4 + §1.6–1.9 are complete as of 2026-04-28. Runbooks in Done. One follow-up below.

1.5 Decide whether to keep the `[coordinator-auth]` diagnostic logs after the soak

Priority: low. **When:** after 2026-05-12 (the scheduled soak-check agent runs that day).

What it is: PR #130 added three `console.log("[coordinator-auth] path=…")` lines to `frontend/lib/coordinatorDashboard.ts::coordinatorHeaders()` that record which auth path was taken on every coordinator-dashboard SSR fetch. They're informational, never log token values, and were load-bearing for the cookie-domain verification on 2026-04-28.

The decision: once the 2026-05-12 soak-check agent confirms three weeks of clean `path=session` lines on prod, the diagnostic value drops sharply. Two options:

- **Keep them long-term** (the code-quality reviewer's default: "safe to leave on permanently"). One-line operator signal whenever the fallback path is hit. Cost is ~3 log lines per coordinator-dashboard request — ignorable.
- **Strip them** and rely on the `ConfigErrorBanner` ops warn (PR #136)
  - the OPERATIONS.md §11.4 runbook for future verification.

Update 2026-04-29: the tier-gate hardening narrowed the soak-check target. Browsers can no longer reach the SSR layer anonymously (middleware redirects pre-render), so `path=none` is unreachable from user traffic — only `path=session` and `path=service-token-fallback` remain. The fallback path still fires on internal server-side calls that don't carry a session cookie. The diagnostic logs are still valid, just less broad in scope. Lean toward "strip" if the soak run shows nothing surprising.

How to verify: the 2026-05-12 agent will open a PR to `docs/handover/auth-soak-check-2026-05-12.md` summarizing the path distribution. Read that PR's findings, then ping me with "keep" or "strip" — either is a 5-minute change.

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2. Data entry (your domain judgment)

> ***Pre-ABQ critical path (in this order):***
>
> 1. ***§2.7 EAZA EEPs first.*** *~30 minutes of admin clicking, two real program rows, immediately moves Panel 2 buckets off zero. Smallest investment, biggest visible-progress signal. Also gives you a rehearsal lap on the admin UI before doing the bulk CARES entry.*
> 2. ***§2.1 CARES population data next.*** *This is the bulk of the work and the load-bearing demo content. If you have a CSV, use the* `seed_populations` *command path (POPULATION_DATA_GUIDE §9) — the manual admin path adds up fast at scale.*
> 3. ***§2.4 CoordinatedProgram CARES rows alongside §2.1.*** *One row per species (not per keeper) so Panel 2 surfaces "CARES priority" at program level. Five minutes per species, can be batched after a few keepers' worth of populations are in.*
> 4. ***§4.1 demo dry-run*** *in the week before ABQ.*
>
> *Everything else (§2.3 / §2.5 / §2.6 / §2.8 / §2.9 / §2.10) is real work but not workshop-critical — chip away post-ABQ or between the above when you need a break from CARES entry.*

2.1 Enter real CARES population data

Priority: HIGHEST — blocks ABQ demo narrative. The coordinator dashboard is a read-only view over this data — without it, ABQ is a demo of empty panels. Step 2 of the critical path above; do §2.7 first for the warm-up lap.

Update 2026-04-30: a demo-shaped seed at `data/seed/coordinator-demo/institutions_populations.csv` is now in the repo. Local Docker has it loaded (9 institutions, 18 populations, 8 programs, 2 transfers, 7 recommendations, 24 events) — staging / prod still need it run there manually:

```
ssh deploy@46.224.196.197
cd /home/deploy/madagascarfish/deploy/staging
docker compose exec -T web python manage.py seed_populations \
    --csv /data/seed/coordinator-demo/institutions_populations.csv
docker compose exec -T web python manage.py seed_demo_coordination
```

The seed is **demo-shaped, not field data** — flagged that way in the README. Real CARES entry below still happens; the seed is a backstop, not a substitute. When you start entering real keepers, delete the matching demo rows or the panels will double-count.

Update 2026-04-29: realistic CARES data scope is "a few keepers with a few cichlid species, almost no other captive populations" — considerably thinner than the original §2.1 framing assumed. That means §2.7 (EAZA EEPs) carries more of the demo narrative weight than originally planned, and Panel 2 buckets will be lopsided toward cichlids. Two implications: (1) prioritize getting §2.7 in even more than before, since it's the ABQ visual-content backstop; (2) the "Coordinator triage view" framing may need to lean on the *structural* story (programs, recommendations, transfer lifecycle) rather than population breadth. Worth thinking about for the §4.1 dry run.

Reduced-scope plan: see `docs/CARES_REDUCED_SCOPE_PLAN.md` — printable runbook covering minimum viable real-data entries (Tier 1 / 2 / 3), interaction with the synthetic `seed_demo_coordination` layer, panel-by-panel expectations with demo talking points, and the honest-vs-padded narrative decision.

Workflow overview: Each CARES hobbyist keeper becomes one `Institution` row. Each population they hold becomes one `ExSituPopulation` row linking species → keeper.

Full walkthrough: `docs/POPULATION_DATA_GUIDE.md` — every field, every decision (e.g. when to check `studbook_managed` for a CARES population — usually unchecked, that's the whole point of the four-bucket split).

If you have a CSV (CARES list in spreadsheet form), skip the manual admin clicking and use the `seed_populations` management command — see `POPULATION_DATA_GUIDE.md` §9. The CSV schema there maps cleanly to a CARES-style export (one row per keeper-species pair). Idempotent, dry-run supported, institutions auto-deduplicated by name.

Concrete steps to start (manual admin path):

1. Sign in to https://api.malagasyfishes.org/admin/ as a Tier 5 (superuser) or Tier 3+ staff account.
2. For each unique CARES keeper on your list:
   - **Populations → Institutions → Add institution**
   - Name: "J. Smith (CARES)" / "Smith household — CARES" / "CARES #4172" — whichever the keeper agreed to. **Pick a convention and keep it consistent across your whole entry session.**
   - Institution type: `hobbyist_keeper` (individual) — not `hobbyist_program`, that's reserved for the CARES org umbrella.
   - Country / city as appropriate (leave city blank for privacy)
   - Leave `contact_email` blank unless the keeper consented (it's Tier 3+ only but still — consent first)
   - Save
3. For each population that keeper holds:
   - **Populations → Ex situ populations → Add**
   - Species: autocomplete, type scientific name
   - Institution: autocomplete, type the keeper's name
   - `count_total` (required-ish), `count_male` / `female` / `unsexed` if known
   - `breeding_status`: `breeding` / `non-breeding` / `unknown`
   - `studbook_managed`: **usually unchecked** for CARES populations
   - `last_census_date`: **set to today** or the date of your CARES data
   - Save
4. After a batch of entries, the admin save hook now auto-triggers "Revalidate public pages" — the public site and dashboard refresh immediately instead of waiting up to an hour for ISR.

How to verify: open `/dashboard/coordinator` — Panel 2 should move species out of "No captive population" into whichever bucket fits (holdings-only, breeding-not-studbook, or studbook-managed).

2.3 Reconcile the 54 unmatched drainage basins

Priority: low-medium. This is a data-quality item — the seed run reports `no drainage basin matched: 54`. Those localities exist but have `drainage_basin = NULL`, so they don't show up on basin-filtered views and can't be grouped by watershed. Not blocking ABQ.

Background: `seed_localities` auto-matches each locality to a HyBAS Level 6 watershed by spatial containment. 54 rows couldn't be matched — usually because the coordinate is marginally outside the basin polygon (coastal points, coordinate imprecision) or the locality is in a basin HyBAS hasn't split out.

Steps:

1. https://api.malagasyfishes.org/admin/species/specieslocality/
2. In the right-hand filter column, check **"Drainage basin: (None)"** and **"Needs review: Yes"** — the rows you care about.
3. For each one, open it and look at the `location` field on the map picker. The map shows the HyBAS watershed polygons as an overlay (if not, zoom in — they can be subtle).
4. If the locality is clearly inside one basin: set `drainage_basin` to that basin via the autocomplete, uncheck `needs_review`, save.
5. If the locality is clearly outside all Madagascar basins (e.g. marine species, coastal estuary): leave `drainage_basin = NULL`, but uncheck `needs_review` with a note in `review_notes` explaining why.
6. If you can't tell: leave it flagged and add a note. Come back later.

How to verify: the count of "needs_review = True" in the list filter drops. Public profiles for those species show their drainage basin on the distribution panel.

Realistic scope: 54 rows is ~30-60 minutes of steady work. Fine to chip at between other tasks.

2.4 Enter CoordinatedProgram rows for known SSPs / EEPs / CARES

Priority: medium-high for ABQ demo narrative.

What it is: `CoordinatedProgram` is the "who runs this species' program" layer (landed in Gate 4 Phase 1). Each row links a species to a formal program — AZA SSP, EAZA EEP, CARES priority, or an independent regional program. Surfaces on Panel 2 (studbook bucketing) and Panel 5 (transfer activity linked to a program).

Steps:

1. https://api.malagasyfishes.org/admin/populations/coordinatedprogram/add/
2. Fill in:
   - **Species**: autocomplete
   - **Program type**: `ssp` / `eep` / `cares` / `independent` / `other`
   - **Name**: human-readable, shown on the dashboard. Suggested format: *"AZA SSP: Madagascar Rainbowfish"*, *"CARES: Paretroplus menarambo"*
   - **Status**: `planning` / `active` / `paused` / `deprecated`
   - **Coordinating institution**: who holds the studbook
   - **Studbook keeper**: the named user (if they're in the system)
   - **Enrolled institutions**: partner zoos / keepers (M2M; dual listbox widget)
   - **Target population size**, **plan summary**, **plan document URL**, **start date**, **next review date** — fill what you know
3. Save.

How to verify: no direct panel for CoordinatedProgram itself yet (Gate 4 Phase 2 candidate). For now, the linkage shows up on Panel 5 when a Transfer references a program.

Workflow tip: if you're entering CARES hobbyist populations and want the species to show up as "CARES priority" at the program level, add **one** CoordinatedProgram row per species with `program_type=cares` — don't create one per keeper.

See also §2.7 below — a starter list of real EEPs you can seed straight from the April 2026 EAZA programme overview.

2.5 Enter Transfer rows as you learn of moves

Priority: medium. Panel 5 ("Transfer activity") renders whatever is here — empty until you start logging.

What it is: `Transfer` tracks animal movement between institutions with a lifecycle (proposed → approved → in_transit → completed / cancelled). Landed in Gate 4 Phase 1.

When to add a Transfer:

- A zoo tells you they're planning to move fish to another institution
- A hobbyist moves founders between keepers
- An accession from the wild (use the collecting institution as source)
- A past move you know happened but isn't recorded

Steps:

1. https://api.malagasyfishes.org/admin/populations/transfer/add/
2. Fill in:
    - **Species / Source institution / Destination institution** (all autocomplete; source must differ from destination — DB-enforced)
    - **Status**: where it is in the lifecycle. `proposed` for something discussed but not agreed; `completed` for a past event.
    - **Proposed date** (required): when it was first logged/discussed
    - **Planned date** (optional): scheduled date
    - **Actual date** (required when status = `completed`)
    - **M.F.U counts** if known (males.females.unsexed)
    - **CITES reference**: permit number for CITES-listed species (CR, EN, VU are all CITES-relevant for Madagascar endemics). Blank for non-CITES cases.
    - **Coordinated program** (optional): if this transfer serves an existing SSP/EEP/CARES program row, link it.
    - **Notes**: quarantine info, holding pens, anything relevant
3. Save. `created_by` auto-fills to you.

How to verify: `/dashboard/coordinator` → Panel 5 "Transfer activity" shows the row in "In flight" (if status is proposed/approved/in_transit) or "Recently completed" (if status= completed and actual_date is within last 90 days).

2.6 Audit existing `studbook_managed` checkboxes

Priority: low. Panel 2's "Studbook-managed" bucket reads this flag directly. If any existing `ExSituPopulation` rows have it set incorrectly (e.g. CARES populations that should be in "breeding_not_studbook"), the bucket counts will be misleading.

Steps:

1. https://api.malagasyfishes.org/admin/populations/exsitupopulation/
2. Filter: **"Studbook managed: Yes"**
3. For each row, ask yourself: "Is there a formal studbook keeper assigned and a coordinated plan?" If no, uncheck.

How to verify: Panel 2 on the coordinator dashboard shows counts that match your mental model.

2.7 Seed real EAZA EEPs from the April 2026 programme overview

Priority: HIGH — start here. These are **real, public** program entries — not demo data — and they make Panel 2 and Panel 5 feel meaningfully populated even before CARES data lands. Step 1 of the critical path

above. ~30 minutes of admin clicking; smallest investment for the biggest immediately-visible dashboard win, and a useful rehearsal lap on the admin UI before §2.1's bulk entry.

Source: `data/reference/April_2026_8e69dc12b4.pdf` (EAZA Ex-situ Programme overview, April 2026, rows 31 and 36 on page 1). These are the only two EEPs that directly touch Madagascar endemic freshwater fish families on the list.

Rows to enter (all via https://api.malagasyfishes.org/admin/populations/coordinatedprogram/add/):

Field	Value #1	Value #2
Species	Pick one Bedotiidae species as the anchor (e.g. *Bedotia madagascariensis*) — or the whole family if we model that later	Pick one Paretroplus species as the anchor (e.g. *Paretroplus menarambo*)
Program type	`eep`	`eep`
Name	`EAZA EEP: Madagascar rainbowfishes (Bedotiidae)`	`EAZA EEP: Cichlids (Cichlidae)`
Status	`active`	`active`
Coordinating institution	Create `Bristol Zoo Project` (type `zoo`, country `United Kingdom`) if not present	Create `National Aquarium Denmark` (type `aquarium`, country `Denmark`)
Studbook keeper	Blank unless Charles Fusari / Peter Petersen get accounts	Blank
Plan summary	"EAZA EEP for the Bedotiidae rainbowfish family. Coordinator: Charles Fusari (Bristol Zoo Project). IUCN: EN. As of April 2026 EAZA overview."	"EAZA EEP for the Cichlidae family. Coordinator: Peter Petersen (National Aquarium Denmark). IUCN: CR. As of April 2026 EAZA overview."
Start date	Blank (not given in the overview)	Blank

Note: Our `CoordinatedProgram.species` is currently a single FK, so "family-level" EEPs need a stand-in species row. Pick the one most representative or that you have the most captive data for. Post-ABQ we can extend the model to carry family-level programs (Gate 4 Phase 2 decision).

Other EAZA EEPs worth noting (from the same overview):

Not Madagascar-endemic but tangentially relevant:

- **Goodeids (Goodeidae)** — Chester Zoo, Joe Chattell, EW-EN. Good reference for CARES-adjacent hobbyist programs; see `Action-Plan-for-the-Conservation-of-Mexicos-Goodeid-Fishes-23-33.pdf`.
- **Sail-fin silversides / Pseudomugilidae** — Jens Bohn, CR-DD.
- **Toothcarps (Valenciidae)** — Bristol Zoo Project, Brian Zimmerman, CR.

These are not Madagascar species, so don't enter them against our registry — but they're useful context if someone at ABQ asks "who else runs freshwater-fish EEPs."

How to verify: `/dashboard/coordinator` Panel 2 shows non-zero "Studbook-managed" or "Breeding, not studbook" counts; Panel 5 "in flight" stays empty until you log Transfers.

2.8 Enter BreedingRecommendation rows

Priority: medium for ABQ demo narrative. Panel 6 on the coordinator dashboard ("Open breeding recommendations") only renders what's in this table — empty until you log.

What it is: `BreedingRecommendation` is the coordinator's to-do list. Landed in Gate 4 Phase 2. Each row captures one recommendation for a species (or a specific population of that species) with a priority, type, and lifecycle status. Shaped against the EAZA Population Management Manual §3.14 three-category cut and the AZA SSP Handbook Chapter 4 Breeding and Transfer Plans structure.

When to add a row:

- A coordinated program (SSP / EEP / CARES) has published a plan and recommends a specific species-level action (breed, hold, transfer).
- A working-group meeting produced an action item with an owner and a due date.
- You as coordinator want to log "this population needs breeding intervention" so it surfaces on the dashboard for follow-up.

Steps:

1. https://api.malagasyfishes.org/admin/populations/breedingrecommendation/add/
2. Fill in the **What** section:
   - **Species**: autocomplete
   - **Recommendation type**: `breed` / `non_breed` / `transfer` / `other`. (Use `transfer` when the plan calls for moving animals; a matching Transfer row in §2.5 is the operational execution.)
   - **Priority**: `critical` / `high` / `medium` / `low`. Critical lights up red on the dashboard; use sparingly.
   - **Rationale**: free text — why this recommendation exists. Copy from SSP/EEP plan, working-group minutes, or your own note.
3. **Who / where** (all optional):
   - **Coordinated program**: link to the SSP/EEP/CARES row this recommendation was issued under, if any (§2.4 entries).
   - **Source population**: the specific `ExSituPopulation` this applies to, when the rec is population-specific (e.g. "breed Bristol's menarambo pair").
   - **Target institution**: who the rec is directed to, when it's directed. Null for coordinator-wide calls.
4. **Lifecycle**:
   - **Status**: start at `open`; flip to `in_progress` when the target institution starts work; `completed` / `superseded` / `cancelled` are terminal.
   - **Issued date**: required. When the recommendation was issued.
   - **Due date**: optional. If set AND in the past, the dashboard tags the row as overdue.
   - **Resolved date** / **outcome notes**: fill in at terminal transition.
5. Save. `issued_by` auto-fills to you on create; `resolved_by` auto-fills when you move status to a terminal state.

How to verify: `/dashboard/coordinator` → Panel 6 "Open breeding recommendations" shows the row sorted by priority (critical first). Completed / superseded / cancelled rows drop out of the panel.

2.9 Log BreedingEvent rows as events happen

Priority: low for ABQ but high ongoing — this is the ledger that makes longitudinal analysis possible post-ABQ.

What it is: `BreedingEvent` is the per-population event log. Landed in Gate 4 Phase 2. One row per discrete event: spawning, hatching, mortality, acquisition, disposition. Count deltas are **signed** — a three-male mortality is `count_delta_male = -3`, a thirty-fry hatch is `count_delta_unsexed = +30`. The table is a ledger; it doesn't auto-update `ExSituPopulation.count_*`.

When to add a row:

- A spawning / hatching event occurs (fish ex-situ is almost exclusively oviparous — log both the spawn and later the hatch as separate rows).
- A mortality event (individual or group).
- Animals arrive from another institution or the wild (acquisition).
- Animals leave (disposition) outside the Transfer lifecycle — e.g. euthanasia, escape, loss to natural cause.

Steps:

1. https://api.malagasyfishes.org/admin/populations/breedingevent/add/
2. Fill in:
    - **Population**: autocomplete (species at institution).
    - **Event type**: the six-value enum above.
    - **Event date**: when it actually happened.
    - **Count deltas** (all signed, all optional):
        - A spawning with no immediate recruit: leave deltas blank.
        - A hatching that produced 30 fry: `count_delta_unsexed = 30`.
        - A mortality of 2 females + 1 male: `count_delta_male = -1`, `count_delta_female = -2`.
        - An acquisition of 5 males + 5 females: `count_delta_male = 5`, `count_delta_female = 5`.
    - **Notes**: anything relevant. Water conditions, clutch size, mortality cause.
3. Save. `reporter` auto-fills to you.

What this does NOT do: the event row is the ledger — it doesn't adjust the running `ExSituPopulation.count_total` / `count_male` / etc. Update those separately in the population's admin page after logging the event. The split is intentional: events are immutable history; counts are current state.

How to verify: open the population in admin (`/admin/populations/exsitupopulation/<id>/`) — the inline at the bottom of the page lists the population's breeding events in reverse-chronological order.

No dashboard panel yet. Events are for the record, not the coordinator triage view. A Gate 4 Phase 3 "recent reproductive activity" panel is plausible once you have enough events logged that it'd read as a trend line instead of a sparse list.

2.11 Approve translated species rows (FR now; DE + ES post-ABQ)

Priority: HIGH — blocks French flag-flip for ABQ. The French machine translation + voice-review pipeline finished 2026-04-30 with 143 species rows sitting at `writer_reviewed`. Until you walk through them and bulk-approve to `human_approved`, the public site continues to fall back to English on `/fr/` URLs (the review-gate is enforced in prod). Same workflow comes back post-ABQ for German + Spanish.

Status table:

Locale	Frontend UI catalog	Species `distribution_narrative`	Backend `gettext .po`	Public flag
`fr`	Translated + voice-reviewed	**143 rows in `writer_reviewed` — your approval pass**	Translated	`..._FR=false` (flip after approval)
`de`	MT pre-staged (786 strings)	143 rows in `mt_draft` (post-ABQ)	Translated	`..._DE=false`
`es`	MT pre-staged (786 strings)	143 rows in `mt_draft` (post-ABQ)	Translated	`..._ES=false`

Two-column comparison — side-by-side admin walkthrough (the operational path):

The TranslationStatus admin's change form puts the English source read-only on top and the target locale below, editable. One row per `(species, field, locale)` tuple — so for a single species, comparing EN vs FR lives in one row, EN vs DE lives in a sibling row, etc.

1. https://api.malagasyfishes.org/admin/i18n/translationstatus/ (or `localhost:8000/admin/i18n/translationstatus/` for local).
2. Right-hand filter column:
   - **Locale** = `fr` (start with French; switch to `de` / `es` post-ABQ for the same flow).
   - **Status** = `writer_reviewed` (these are the rows that both the MT pipeline and the conservation-writer agent already pre-screened).
   - **Content type** = `Species`.
3. **Sort by `object_id`** (the column header). This groups rows by species — useful because each species can have up to 4 translatable fields (`distribution_narrative` , `description` , `ecology_notes` , `morphology`). Most species today only have `distribution_narrative` populated, so it's mostly one row per species.
4. Click into a row. The change form shows:
   - **English source** pane (read-only, gray background) — the authoritative EN text from `Species.<field>_en`.
   - **Target translation** pane (editable, yellow background) — the current FR (or DE / ES) text from `Species.<field>_<locale>`.
5. Read both panes. Three outcomes:
   - **Approve as-is**: hit Save (no edits), then bulk-approve in step 6 below.
   - **Edit then approve**: tweak the target pane, hit Save (writes back to `<field>_<locale>` on the species row), then bulk-approve.
   - **Send back to mt_draft**: substantive concern → bulk-action "Send back to mt_draft" demotes the row; conservation-writer can re-review.
6. **Bulk approve a batch:** back on the list view, multi-select reviewed rows, Action dropdown → **"Approve: writer_reviewed → human_approved"** → Go. Stamps `human_approved_by` (= you) and `human_approved_at` (= now) on every selected row.

Three-locale comparison — CLI for spot-checking one species across EN / FR / DE / ES:

The admin only shows EN vs ONE target locale at a time. To eyeball all four locales side-by-side for one species (good for catching MT drift between locales — e.g. FR says "rivière X" but DE invented "Fluss Y"), use the public API. No auth needed; `Accept-Language` toggles which locale the response field comes back in:

```
SP_ID=1   # Bedotia albomarginata; pick whichever species you're checking

for LOCALE in en fr de es; do
  echo "=== $LOCALE ==="
  curl -s -H "Accept-Language: $LOCALE" \
       "http://localhost:8000/api/v1/species/$SP_ID/" \
       | python3 -c "import sys,json; d=json.load(sys.stdin);
print(d['distribution_narrative'])"
  echo
done
```

That prints the English source plus the three translated versions back to back. If the locale fallback badge fires for any of them (`<field>_locale_actual = 'en'` even though you asked for `fr`), the review-gate hasn't approved that row yet — see step 6 above.

Direct DB comparison — if you want raw `_en` / `_fr` / `_de` / `_es` columns for one species ignoring the review-gate, drop into the Django shell:

```
docker compose exec -T web python manage.py shell -c "
from species.models import Species
sp = Species.objects.get(pk=1)
for f in ('description', 'ecology_notes', 'distribution_narrative', 'morphology'):
    print(f'--- {f} ---')
    for loc in ('en', 'fr', 'de', 'es'):
        val = getattr(sp, f'{f}_{loc}', '') or ''
        print(f'  {loc}: {val[:200]}')
    print()
"
```

FR approval pass — your TODO this week:

- [ ] Filter to `locale=fr` , `status=writer_reviewed` , `content_type=Species` (143 rows expected).
- [ ] Walk family by family. **Bedotiidae (29) first** — that's the ABQ banner family and the smallest meaningful batch. Then Cichlidae (48 — biggest), Aplocheilidae (8), Anchariidae (6), and the 21 small families (~52 rows combined).
- [ ] Bulk-approve each family's batch via the admin action.
- [ ] When at least Bedotiidae is approved, run the flag-flip pre-flight in `docs/operations/i18n-flag-flip-runbook.md` and flip `NEXT_PUBLIC_FEATURE_I18N_FR=true` in Vercel production.
- [ ] Smoke-test `/fr/species/<bedotiidae-id>` on prod — French distribution narrative should render (no `(English)` fallback badge on the approved fields).

DE + ES — post-ABQ: same flow, swap `fr` → `de` / `es` in the filter. No human-approved rows yet, so the public flag stays `false` . See `docs/planning/specs/gate-L5-german.md` and `docs/planning/specs/gate-L6-spanish.md` for the post-ABQ workplan (plurals + voice review + family-batched approval, ~1 day per locale).

How to verify: run the multi-locale CLI snippet above against an approved species and confirm the FR field renders the French text (not the English fallback). On prod with the gate on, only `human_approved` rows

make it through; on local with `I18N_ENFORCE_REVIEW_GATE=False`, `writer_reviewed` rows also render (useful for testing the comparison flow before flipping anything).

2.10 Reconcile SHOAL priority flags using the worklist report

Priority: medium. Not blocking ABQ, but gets the public species directory and dashboard captions accurate about which species carry SHOAL's priority flag.

What it is: a read-only management command (`shoal_priority_report`) that diffs `data/reference/ActiveShoalPriorityMadagascar.csv` against the registry and prints three lists. It makes no writes — you use the output as a worklist to manually flip `Species.shoal_priority` in admin, reviewing each flip against the current SHOAL website or published list.

Why the manual gate: the CSV has no version / provenance metadata, taxonomy drift (`Pachypanchax omalonota` vs `omalonotus`) will cause false negatives on direct string match, and 8 species currently flagged in the registry aren't in the CSV. An auto-sync risks silently un-flagging species that should stay flagged.

Run it (on staging or locally):

```
docker compose exec -T web python manage.py shoal_priority_report \
  --csv /data/reference/ActiveShoalPriorityMadagascar.csv
```

The output has three sections:

- **In CSV, not flagged True in registry** — candidates to flip True.
- **Flagged True in registry, not in CSV** — candidates to review (flip False? or CSV is stale?).
- **In CSV, not in registry at all** — likely taxonomy drift or missing species in our seed; investigate before adding.

Workflow:

1. Run the command. Save the output.
2. For each entry in list #1: open `/admin/species/species/?q=<scientific name>` and flip `shoal_priority = True` if SHOAL's current website still lists them.
3. For each entry in list #2: check if SHOAL removed them in a newer list. If yes, flip False. If unclear, leave as-is.
4. For each entry in list #3: check taxonomy (is it a synonym of something we have?) — the `Pachypanchax omalonota` vs `omalonotus` pattern is the most likely culprit. If it's a real missing species, it should be added via the species seed process, not this command.

When to re-run: when SHOAL publishes an updated priority list. Annually at most.

---

3. Reference documents

3.1 Received and parsed (April 2026 batch)

The following arrived in `data/reference/` on 2026-04-23 and have been skimmed. Each closes a previous "need" item in this section.

File	What it is	What it unlocks

`April_2026_8e69dc12b4.pdf`	EAZA Ex-situ Programme overview, April 2026 — the live registry of every EEP with coordinator + institution + IUCN status	**§2.7 above** — two concr EEPs to seed right now (Bedotiidae, Cichlidae). Supersedes the §3.2 ask
`aza_ssp_handbook.pdf`	AZA Species Survival Plan Program Handbook (2024 revision)	Settles §3.1. Phase 2 mo shape for `BreedingRecommendation` w follow the "Breeding and Transfer Plan" structure from Chapter 4; sustainability designatior (green/yellow/red SSP) informs a future enum.
`EAZA_Population_Management_Manual_V6_2_67db7a6627.pdf`	EAZA Population Management Manual V6.2 (Jan 2025)	Settles §3.2. Introduces TAG → EEP → Species Committee hierarchy anc the Long-Term Management Plan (LTMP concept; also distinguish EEP (intensive) from ESB (European Studbook, les intensive). Phase 2 may add `program_type='esb'` a an optional TAG layer abc `CoordinatedProgram`.
`starting_a_new_aza_studbook_in_zims_for_studbooks_2023.pdf`	ZIMS for Studbooks — SSP startup workflow	Partial for §3.3. Covers th operator workflow, not th field-level schema.
`how_to_publish_a_studbook_using_zims_for_studbooks_mac.pdf`	ZIMS for Studbooks — publication workflow	Same as above.
`guidelines_for_data_entry_and_maintenance_1996.pdf`	Studbook data entry guidelines	Dated but useful — the p animal field list is

	(1996, SPARKS era)	essentially stable. Inform `BreedingEvent` shape.
`Action-Plan-for-the-Conservation-of-Mexicos-Goodeid-Fishes-23-33.pdf`	Goodeid Action Plan 2023-2033	Domain analog — Mexica Goodeids are a parallel hobbyist-heavy CARES-adjacent case. Useful for workshop narrative but doesn't change the data model.
`ActiveShoalPriorityMadagascar.csv`	SHOAL Priority Species — active Madagascar subset (44 rows)	Settles §3.6. A `reconcile_shoal_priority` management command c diff this against the regis and flip `Species.shoal_priority=T` for matches. Good half-c task when you want to ru it.

What this let me do: Gate 4 Phase 2 landed on 2026-04-23 with `BreedingRecommendation` + `BreedingEvent` models shaped against the AZA Breeding and Transfer Plan structure (Chapter 4 of the SSP Handbook) and the EAZA annual recommendations categories (breed / non-breed / transfer, per §3.14 of the EAZA manual). Data entry workflows for both are in §2.8 and §2.9 above.

3.2 Still outstanding

Ask	Priority	Notes
Current CARES priority species list — CSV	high	The EAZA overview isn't this. Need the authoritative CARES list as a CSV (or URL) so I can diff against `Species.cares_status` and reconcile. If you have access to the CARES database or maintainer, request.
PMx / PMxWeb input format	low	Only matters if we want the platform to export directly into PMx for demographic analysis. Not blocking ABQ or Phase 2.
ZIMS field dictionary (not just workflow)	low-medium	The two ZIMS PDFs cover the operator workflow. A per-module field schema (CSV export column list or PDF dictionary) would let us shape models to be ZIMS-exportable in the future. Nice to have, not essential.

---

4. Workshop prep (ABQ, June 1-5, 2026)

4.1 Dry-run the coordinator dashboard demo

Priority: high, closer to event.

Steps: walk through `/dashboard/coordinator` end-to-end with real seeded data, ideally in the week before ABQ. Check:

- All five panels populated and telling a coherent story (Panel 5 is Transfer activity; don't forget it)

- "Show N more" on Panel 1 works with realistic list lengths
- Tier 3+ gating works (log in as a non-admin tier 3 user too)
- M.F.U formatting on Panel 3 reads sensibly with real ratios
- Panel 4 catches populations you know are overdue
- Coordinator nav link highlights correctly when you're on the page (and doesn't double-highlight with Dashboard)

How to verify: the person demoing (presumably you) can talk through each panel without awkward empty states or visible bugs.

4.2 Decide whether to keep the Coordinator nav link visible at ABQ

Priority: low. The link is live as of PR #102 — anyone on the site sees it. You can still hide it before the demo if the "invited to see this" framing matters more than discoverability.

If you want to hide it for ABQ: ping me and I'll drop the nav entry on a short branch; can put it back after the workshop. Keeping it visible is the default.

Update 2026-04-28: with `NEXT_PUBLIC_FEATURE_AUTH=true` now flipped on, the nav also shows Sign in / Sign up to anonymous visitors. Same question applies — keep or hide for ABQ. Default is keep.

Update 2026-04-29: mostly self-resolves now that the Coordinator nav link auto-hides for anonymous + Tier 1/2 viewers (see latest tier-gate fix). Anyone visiting without a Tier 3+ session never sees the link. Remaining open question is narrower: while you're demoing *signed in as Tier 3+*, the link IS visible — does that matter for the "invited to see this" framing with attendees looking over your shoulder? Default still: keep.

4.3 Audit test-data hygiene before the demo

Priority: medium, do in the week before ABQ.

What to check:

1. **Seeded e2e users.** The Playwright workflow (`frontend-auth-e2e.yml`) creates `researcher-e2e@example.com` (Tier 2), `coordinator-e2e@example.com` (Tier 3, member of "E2E Test Zoo"), and `admin-e2e@example.com` (Tier 5) via the `seed_test_users` management command. **These exist on prod too** if the Hetzner deploy ever ran the seed. Audit: `<https://api.malagasyfishes.org/admin/accounts/user/?q=e2e>` — if any rows turn up, decide whether to leave them (harmless test accounts with predictable passwords) or delete (cleaner demo-state).
2. **The "E2E Test Zoo" institution.** Check `<https://api.malagasyfishes.org/admin/populations/institution/?q=E2E>` . If present in prod and you don't want it visible during the demo (it'd appear in any institution-level panel that lists test rows), delete it.
3. **Your real-name Tier 3 user from the cookie-domain check.** You created a Tier 3 account on prod on 2026-04-28 to verify the session-based SSR forwarding (`[coordinator-auth] path=session`). Decide: leave it (you'll demo from this account at ABQ), rename it for clarity, or remove and re-create later.

How to verify: Django admin user list shows only the accounts you intend to demo from. Coordinator dashboard panels don't show "E2E Test Zoo" rows.

4.4 Post-ABQ — fix top nav responsive wrap (deferred)

Priority: post-workshop polish, not a blocker. Diagnosed by ux-reviewer 2026-04-28.

What's wrong: when the viewport is narrower than ~960px (most tablets, all phones), nav items in `frontend/components/NavLinks.tsx` wrap to a second row that renders **outside** the fixed-height header band in `frontend/components/SiteHeader.tsx` (`height: 72`). Items are still in the DOM and tab-focusable, just visually clipped behind the sticky-header backdrop and the page content scrolling under it. Accessibility smell on top of the layout bug.

Why deferred: ABQ demo runs on a projector at desktop width (≥1280px), where all 7 items fit comfortably. The realistic phone scenario is "stakeholder pulls site up at the coffee break" — fix it in the polish window after the workshop.

Heads up during the demo: don't whip out your phone mid-demo and scroll the nav.

When you pick this up post-ABQ: ux-reviewer enumerated 7 patterns (hamburger, overflow-into-More dropdown, horizontal scroll, hide-by- priority, multi-row auto-grow header, icon-only collapse, two-tier nav). Quick win is changing `height: 72` to `min-height: 72` in `SiteHeader.tsx` — honest visual, eats some vertical space, no JS changes. Pick a real pattern based on whether mobile is a real persona for the platform or a courtesy.

---

5. Things I explicitly haven't asked you to do

Listing these so you know they're not on your plate.

- **Backend code**: I write and test all of it in Docker locally.
- **Frontend code**: same — you don't need Node/pnpm running.
- **Migrations**: I generate them locally; they apply on staging automatically via the `deploy-staging` GitHub Actions workflow on every merge. You don't need to run `migrate` by hand unless the deploy workflow fails silently (OPERATIONS.md §12.6).
- **Running the seed**: ditto — the deploy action won't re-seed on every merge, but when a seed PR lands you'll see it called out in the PR body.
- **Writing tests**: I write the unit and integration tests.

Done

Completed items are archived here. We keep the full how-to for each so there's a runbook to refer back to when something needs to be re-done (e.g. secret rotation). Newest at the top.

1.9 Verify the cookie-domain check on production — DONE

Completed: 2026-04-28. **Unlocks:** confirms Tier 3+ users on prod actually use their session token for SSR fetches, not the `COORDINATOR_API_TOKEN` fallback. Without this check, the architectural goal of Gate 11 ("no service tokens mailed around") could be silently undermined — every signed-in user would be anonymously bypassing tier without anyone realizing.

The verification: signed in as the production Tier 3 user, hit `/dashboard/coordinator`, and read Vercel function logs for the `[coordinator-auth]` lines (PR #130 instrumentation in `frontend/lib/coordinatorDashboard.ts::coordinatorHeaders()`). Result: every recent line showed `path=session`. No `path=service-token-fallback` or `path=none` regressions.

Soak follow-up: scheduled remote agent fires on 2026-05-12 to re-run the same check and post a summary PR to `docs/handover/auth-soak-check-2026-05-12.md`. See §1.5 active item for the post-soak diagnostic-log retention decision.

Full runbook: `OPERATIONS.md §11.4`.

1.8 Set Preview-scope `NEXTAUTH_SECRET` + `NEXTAUTH_URL` in Vercel — DONE

Completed: 2026-04-28. **Unlocks:** preview deploys (`*.vercel.app` PR previews) no longer emit `[next-auth][error][NO_SECRET]` 500s on `/api/auth/session` from the client-side `useSession()` poll. Real auth flows still don't complete on preview because of the cookie-domain mismatch (architecture §9) — that's intentional. These env vars exist solely to silence the error.

Values used:

- `NEXTAUTH_SECRET` (Preview scope) = fresh `openssl rand -hex 32`, distinct from the prod secret.
- `NEXTAUTH_URL` (Preview scope) = placeholder string OR `https://${VERCEL_URL}` to use Vercel's dynamic preview hostname (either silences the error equally well).

On rotation: mint a new hex, paste into Preview scope, save. No redeploy needed on already-running previews — but the next preview build will pick it up.

1.7 Flip `NEXT_PUBLIC_FEATURE_AUTH=true` in Vercel prod — DONE

Completed: 2026-04-28. **Unlocks:** Gate 11 auth surface is now live on prod. Login / Sign up nav appears for anonymous visitors; Account / Sign out for signed-in. Middleware redirects on `/account`. `/dashboard/coordinator` continues to render via the service-token fallback for anonymous SSR (no behavior regression).

Runbook (rollback if regression):

1. Vercel → Settings → Environment Variables → edit `NEXT_PUBLIC_FEATURE_AUTH`.
2. Change value to `false` (or delete the row).
3. Save → redeploy.

After rollback: nav loses Login/Sign-up/Account/Sign-out; middleware becomes pass-through; `/login` `/signup` `/verify` `/account` URLs still resolve for UAT but aren't advertised. Anonymous public surface is unchanged.

On re-enable: set value back to `true`, redeploy. No state migration required — flag is purely a UI gate.

1.6 Add `TRUST_X_FORWARDED_FOR=True` to Hetzner staging `.env` — DONE

Completed: 2026-04-28. **Unlocks:** the per-IP rate limiter on `/api/v1/auth/login/` now sees the real client IP (forwarded by Caddy) instead of Caddy's container IP. Without this, the limiter would either rate-limit the proxy itself (blocking everyone) or — if Caddy's IP varied — fail to throttle at all. The `>=5` boundary fix (PR #132) is dependent on this env var to function correctly behind the reverse proxy.

Runbook (re-do or rotation):

```
ssh deploy@46.224.196.197
cd /home/deploy/madagascarfish/deploy/staging
nano .env
# Add or update:
TRUST_X_FORWARDED_FOR=True
docker compose up -d --force-recreate web
```

Verification: from your laptop, run 5 failed login attempts in a row to a non-existent account; expect `401 401 401 401 429`. The window resets after 15 minutes.

```
for i in 1 2 3 4 5; do curl -s -o /dev/null -w "attempt $i: %{http_code}\n" -X POST
https://api.malagasyfishes.org/api/v1/auth/login/ -H "Content-Type:
application/json" -d '{"email":"nope@example.com","password":"wrong"}'; done
```

On adding a non-Caddy backend: if Django ever gets exposed directly (no trusted reverse proxy in front), set this back to `False` — otherwise an attacker can spoof XFF and rotate the asserted IP per request to bypass the rate limit. The setting is in `backend/config/settings/base.py`; default is `False`.

1.4 Pick an email-deliverability vendor — DONE

Completed: 2026-04-28. Vendor: **Resend** (SMTP). **Unlocks:** Django `send_mail` reaches real inboxes from staging — the `/signup → /verify` email loop is now end-to-end on `api.malagasyfishes.org`. PR #122 wired the SMTP env-var reads in `backend/config/settings/base.py` and refreshed `deploy/staging/.env.example` to mirror what's on the Hetzner box.

Resend SMTP credentials on Hetzner staging `.env`:

```
EMAIL_BACKEND=django.core.mail.backends.smtp.EmailBackend
EMAIL_HOST=smtp.resend.com
EMAIL_PORT=587
EMAIL_HOST_USER=resend
EMAIL_HOST_PASSWORD=<Resend API key, re_...>
EMAIL_USE_TLS=True
DEFAULT_FROM_EMAIL=noreply@malagasyfishes.org
```

`EMAIL_HOST_USER` is the literal string `resend` — that's Resend's convention. The API key is the password.

Smoke-test command (run on the Hetzner box):

```
ssh deploy@46.224.196.197
docker compose -f /home/deploy/madagascarfish/deploy/staging/docker-compose.yml \
  exec -T web python manage.py shell -c \
  "from django.core.mail import send_mail; send_mail('Resend test', 'hello',
'noreply@malagasyfishes.org', ['alekseisaunders@gmail.com'])"
```

Expect inbox delivery within ~30s. Spam landing → re-check SPF/DKIM in DNS. SMTP auth failure → `EMAIL_HOST_PASSWORD` doesn't match the live Resend API key.

Rotation runbook: mint a new Resend API key in the Resend dashboard, update `EMAIL_HOST_PASSWORD` on the Hetzner `.env`, then `docker compose up -d --force-recreate web`. Old key can be deleted once the new one is live.

Domain notes: `malagasyfishes.org` is verified in Resend with DKIM

- SPF records on Cloudflare. If the sending domain ever changes, the DNS records have to be regenerated in the Resend dashboard.

1.3 Generate `NEXTAUTH_SECRET` per environment — DONE

Completed: 2026-04-28. **Unlocks:** NextAuth JWT signing on dev + prod. Cookies issue cleanly with `HttpOnly; Secure; SameSite=Lax` ; no warnings in Vercel function logs. Staging frontend wiring deferred (no `staging` git branch yet — see §1.4 of the Gate 11 auth handover for context).

Runbook (rotation):

1. Generate fresh values on your laptop:

   ```
   openssl rand -hex 32  # one per environment
   ```

   Use a different secret per env. Never reuse.

2. **Dev (`frontend/.env.local`):**

   ```
   NEXTAUTH_SECRET=<dev-hex>
   NEXTAUTH_URL=http://localhost:3000
   ```

3. **Vercel — production:**

   - Project Settings → Environment Variables
   - `NEXTAUTH_SECRET = <prod-hex>` , scope to **Production**
   - `NEXTAUTH_URL = https://malagasyfishes.org` , scope to **Production**
   - Save, redeploy.

4. **Vercel — staging:** not configured. The architecture spec called for a `staging` git branch with `staging.malagasyfishes.org` as a Vercel alias, but the staging frontend was never wired. Today `staging.malagasyfishes.org` is unused (the backend lives at `api.malagasyfishes.org`). Generate a third secret only when the staging frontend is actually stood up.

Verification: browse `/login` while logged out, complete a signup

- verify + login flow on prod — no NextAuth warnings in Vercel logs, session cookie present in browser dev-tools with the expected flags.

On a leak: generate a fresh secret on the affected env, paste into Vercel env vars, redeploy. Existing sessions are immediately invalidated (JWTs signed with the old secret won't decode). Acceptable cost.

1.2 Verify the revalidate webhook is configured — DONE

Completed: 2026-04-22. **Unlocks:** admin saves on Species / Genus / Institution / ExSituPopulation / SiteMapAsset immediately refresh the public site and coordinator dashboard; no more 1h ISR stale window.

Runbook in case of secret rotation (same pattern as 1.1, different names):

- **Vercel env:** `REVALIDATE_SECRET = <secret>`
- **Staging `.env` :**

  ```
  NEXT_REVALIDATE_URL=https://malagasyfishes.org/api/revalidate
  NEXT_REVALIDATE_SECRET=<same value as Vercel's REVALIDATE_SECRET>
  ```

- `docker compose up -d --force-recreate web`

Verification: edit any species in admin and save. The green banner should read "Revalidated N path(s)." If it says "Revalidate is not configured..." the backend env vars aren't set or the container wasn't recreated. OPERATIONS.md §11.1.

1.1 Set `COORDINATOR_API_TOKEN` on Vercel + staging backend — DONE

Completed: 2026-04-22. **Unlocks:** `/dashboard/coordinator` renders panel data from the Tier 3+ API endpoints without a user session (service-token SSR).

Runbook in case of secret rotation:

1. Generate a secret on your laptop:

   ```
   openssl rand -hex 32
   ```

2. **Vercel** (Project Settings → Environment Variables):
   - Key: `COORDINATOR_API_TOKEN`
   - Value: the hex string
   - Environments: **Production** (+ Preview if you want preview branches to fetch coordinator data)
   - Save, then redeploy the project.

3. **Staging backend** — SSH in and edit the compose `.env`:

   ```
   ssh deploy@46.224.196.197
   cd /home/deploy/madagascarfish/deploy/staging
   nano .env
   ```

   Add / update:

   ```
   COORDINATOR_API_TOKEN=<same hex string>
   ```

   Save, then recreate the web container:

   ```
   docker compose up -d --force-recreate web
   ```

Verification:

```
curl -s -o /dev/null -w "%{http_code}\n" \
  -H "Authorization: Bearer <your token>" \
  https://api.malagasyfishes.org/api/v1/coordinator-dashboard/stale-census/
```

Should print `200`. Without the header it should print `403`.

Full runbook: OPERATIONS.md §11.2.

2.2 Re-upload silhouettes for Paretroplus menarambo + P. nourissati — DONE

Completed: 2026-04-22. **Unlocks:** species-specific silhouettes render on the directory card and profile page for both species; no more placeholder fish.

The admin SVG file picker (PR #86) made this a drag-and-drop: open the species in `/admin/species/species/`, use the "Upload .svg file (optional)" field in the Silhouette section, save.

3.1 / 3.2 / 3.3 (partial) / 3.6 reference documents — DONE on 2026-04-23

See §3.1 above for the received-docs table. Remaining outstanding asks moved to §3.2.